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7.1 INTRODUCTION

7.1.1 This Addendum to Chapter 7: Noise and Vibration has been prepared by Brookbanks Consultants Ltd in response to the request that further consideration should be given to the need to ensure that all existing noise sources, including existing industrial activities at Thornhill Industrial Estate and SDC-Keypoint, have been assessed including any mitigation measures that may be appropriate insofar as they might be relevant.

7.1.2 This assessment provides additional information to the assessment that was presented in the original ES (October 2013) and as such the impacts in relation to noise and vibration during both the construction and post-completion stages remain as presented in the original ES.

7.2 ASSESSMENT METHODOLOGY

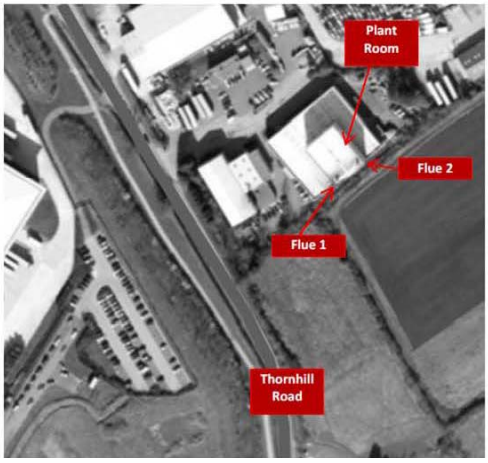
Scope

7.2.1 For the purposes of this assessment, the Study Area has been taken as the Application Site. The scope of the assessment includes consideration of noise sources potentially associated with Keypoint. Such sources could potentially comprise noise from HGVs and with the use of the railway for freight movement. Such activity is remote from the application site and generally shielded from the development by the existing SDC-Keypoint buildings and existing 4.5 metre bund established to the west of Thornhill Road.

7.2.2 With regard to the Thornhill Industrial Estate, although containing a number of businesses, the level of noise generating activity is generally low. Indeed it is characteristic of the area that the noise environment on the site and its vicinity is characterised by traffic related noise rather than other sources. This was highlighted in the original ES assessment and remains the case.

7.2.3 Within the Thornhill Industrial Estates is located a premises occupied by Synergy Health with noise generated from a ground floor plant room and two flues located on

the roof of the building. The flues on the roof of the building operate on an intermittent and occasional basis.



Assessment approach

7.2.4 Methods of assessment have been employed that are consistent with current guidance and recommendations in the form of statutory documents and recognised publications to ensure that the findings are robust.

7.3 RELEVANT POLICY

British Standard 8233:2014; Sound Insulation and Noise Reduction for Buildings

7.3.1 BS8233:2014[1] gives recommendations for the control of noise in and around buildings and provides guidance in relation to criteria and internal noise levels for habitable rooms of residential dwellings. The following internal and daytime noise targets are suggested within BS8233 to be met within sensitive rooms of the residential dwellings:

35dB LAeq (16 hour) during the daytime in living rooms;

40dB LAeq (16 hour) during the daytime in dining rooms

30dB LAeq (8 hour) during the night time in bedroom areas.

[1] British Standard 8233:2014; Sound Insulation and Noise Reduction for Buildings

55 dB LAeq,(16 hour) for noise in external amenity areas.

7.3.2 It is recognised that such target values will not always be achievable in practice for instance where dwellings are located close to busy roads.

7.3.3 That a rigid requirement is not anticipated is reflected in the approach set out in the Noise Policy Statement for England (2010) which introduced the concept of more flexible measures of Significant Observed Adverse Effects.

7.4 BASELINE CONDITIONS

Noise Monitoring

7.4.1 Additional existing noise measurements have been carried out on the application site adjacent to the Thornhill Industrial Estates including the building presently occupied by Synergy Health. A background survey has been carried out in order to assess the existing noise levels on an area of land to the east of Thornhill Road, South Marston, Swindon. The background survey was conducted at two monitoring locations covering a typical 24 hour period on Tuesday 10th November 2015.

7.4.2 The monitoring locations are identified below.

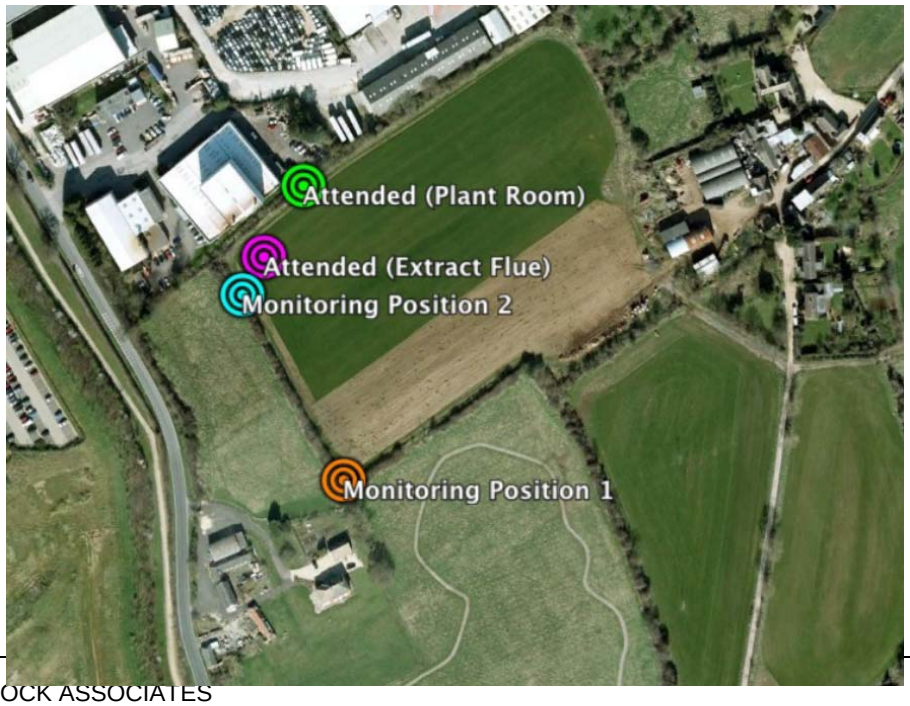


Figure 7.1: Noise Monitoring Locations

7.4.3 All acoustic measurement equipment used during the noise surveys conformed to Type 1 specification of British Standard 61672: 2003: Electroacoustics, Part 1 Specifications.

Table 7.2: Noise Monitoring Equipment Inventory

Equipment Description	Manufacturer	Serial number
Sound Level Meter	**Norsonic 118**	28952
Sound Level Meter	**Norsonic 118**	30559
Sound Level Meter	**Cassella CEL 480**	089653
Acoustic Calibrator	**Norsonic 1251**	32856

7.4.4 The results of the noise survey are indicated below.

Table 7.3: Monitoring results range

Criteria	Monitoring Position 1		Monitoring Position 2	
	Day (07:00 to 23:00)	Night (23:00 to 07:00)	Day (07:00 to 23:00)	Night (23:00 to 07:00)
$LA_{eq\ 1\ hour}$	46 - 53 dB	46 – 49 dB	43 – 56 dB	43 – 55 dB
$LA_{eq\ 16\ hr\ /\ 8\ hr}$	52 dB	48 dB	54 dB	51 dB
$LA_{90\ 1\ hour}$	43 – 51 dB	43 – 45 dB	44 – 55 dB	41 – 53 dB
$LA_{10\ 1\ hour}$	48 - 54 dB	48 – 50 dB	48 - 58 dB	45 – 56 dB
$LA_{max\ I\ hour}$	60 – 72 dB	57 – 63 dB	57 – 73 dB	56 – 65 dB

7.4.5 The background surveys incorporated noise from the adjacent Thornhill Industrial Estate including the property presently occupied by Synergy Health and from the Keypoint and Honda developments.

7.4.6 Predominantly the background noise is characterised by road traffic in the area. Separate monitoring in respect of an application for development at Crown Timber noted that HGV movements were audible during monitoring but did not affect measured noise levels.

Table 7.4: Attended monitoring results range

Critera	Attended (Flue)		Attended (Plant room)
	Day (07:00 to 23:00)	Night (23:00 to 07:00)	Night (23:00 to 07:00
$LA_{eq\ 5\ min}$	58 – 59 dB	57 – 58 dB	61 – 62 dB
$LA_{90\ 5\ min}$	57 – 58 dB	56 – 58 dB	61 – 62 dB
$LA_{10\ 5\ min}$	59 – 61 dB	58 – 63 dB	61 – 63 dB
$LA_{max\ 5\ min}$	60 – 64 dB	60 – 67 dB	64 – 67 dB

7.4.7 Table 7.4 used attended surveys to capture noise events from the extract flue and plant room. Measurements were taken on Friday 6th November 2015 and Tuesday 10th November.

7.4.8 The attended surveys also suggest that the noise levels associated with the plant room (at ground floor level) slightly exceed and/or last longer than those associated with the more occasional flues. Noise from the plant room, at ground floor level, would be capable of mitigation through barriers at ground floor level.

7.5 POTENTIAL EFFECTS

Operational Impacts: British Standard 8233:2014; Guidance on Sound insulation and noise reduction for buildings

7.5.1 The noise survey commissioned in November 2015 included the northern boundary of the site adjacent to the Thornhill Industrial Estate which reflects the closest potential receptor.

7.5.2 Guidance in respect of potential noise targets in BS8233 is set out in section 7.3 above.

External Noise

The free field noise levels closest to the Thornhill Road Industrial Estate during the daytime are measured as 54 dB (Table 7.3) and therefore are already below the suggested desirable target of 55 dB LAeq,(16 hour) for noise in external amenity areas. The experience of noise in gardens and amenity areas will be less than that presently observed as free field noise levels as a consequence of the natural

screening provided by property boundaries, the layout of buildings in respect of garden space and the additional distance of amenity areas from the industrial estate.

Façade Noise on Properties adjacent to Thornhill Industrial Estate

7.5.3 In assessing noise levels at the facades of properties consideration needs to be given to noise levels both at day and night. At the façade, account is also taken of the possible additional effects of reverberation of noise to allow for a conservative or precautionary assessment. The free field noise levels during the daytime and night are measured as 54 dB and 51 dB respectively (Table 7.3). A correction of 3dB has been included to allow for any effects of reverberation.

7.5.4 As a consequence potential noise levels at the façade are 57 and 54db during the daytime and nighttime respectively. Substantial attenuation of such external façade noise is provided by the windows of each property. Attenuation of external to internal noise levels provided by a window partially open for ventilation is 15dB. Standard double glazing specification consists of panes that are 6mm wide separated by a 12mm gap, when fully closed can attenuate 33 dB.

7.5.5 Therefore internal noise levels will reduce to 24dB and 21dB during the day and night time respectively when taking into account the 33dB sound attenuation provided by a closed window. Therefore, this demonstrates that when windows are closed, the internal noise levels set out as targets in BS8233 are achieved, irrespective of the location of housing in respect of the Thornhill Industrial Estate and the building presently occupied by the Synergy Heath facility.

7.6 MITIGATION MEASURES

External Noise Standards

7.6.1 The recorded LAeq 16 hour is recorded at 54 dB. This demonstrates that the external limit can be achieved without mitigation.

7.6.2 Furthermore, the buildings and layout will provide further natural screening to the garden areas which will result in any event in lower noise levels being achieved.

Direct façade noise levels on the proposed dwellings

Internal Noise Standard – Day Time Habitable rooms

7.6.3 The external LAeq 16 hour day time is identified at 57 dB (at a location adjacent to the Industrial Estate), when including an increase in noise due to the façade effect (+3dB). Standard double glazing can attenuate sounds levels by 33 dB when closed, with 15 dB reduction through an open window. A habitable room is classified as a living /dining room.

7.6.4 This will reduce the internal noise level to 24 dB (closed windows) and 42 dB (open window). This demonstrates that BS8233 target internal noise levels can be achieved with closed windows without mitigation.

7.6.5 Mitigation could be considered to achieve such internal standards with open windows, although it is recognised that such target values need not always be achievable in practice for instance where dwellings are located close to busy roads.

7.6.6 The layout of the Proposed Development and the internal arrangements of properties will be subject to further detailed design. Through the design process, consideration may be given to orientating those dwellings closest to the Thornhill Road Industrial Estate, such that habitable rooms are not facing the Estate

7.6.7 Should any habitable rooms be facing the Estate facility, then additional screening or alternative ventilation could be considered. The latter can be achieved through trickle vents / air bricks

Internal Noise Standard – Night Time Habitable rooms

7.6.8 The potential internal LAeq 8 hour night time at night time(at a location adjacent to the Industrial Estate) is calculated from the external noise levels identified at 54 dB, when including an increase in noise due to the façade effect. Standard double glazing can attenuate sounds levels by 33 dB when closed, with 15 dB reduction through an open window.

7.6.9 This will reduce the internal noise level to 21 dB and 39 dB respectively and demonstrates that BS8233 internal noise level targets can be readily achieved with closed windows.

7.6.10 Should any habitable rooms be facing the Industrial Estate facility, then additional screening or alternative ventilation could be considered for those properties closest to the Industrial Estate the latter can be achieved through trickle vents / air bricks.

Mitigation Extent

7.6.11 Typically, the first row of housing fronting onto the noise source experiences the highest noise levels. Adjacent dwellings behind are screened by the first row of housing. Any mitigation that may be considered necessary can be achieved by barriers to noise eg. in relation to the plant room but may be less effective in relation to the flues due to their height. The assessment indicated that the internal noise levels would be 42 dB and 39 dB during the day and night time respectively with open windows –adjacent to the Industrial Estate. With closed windows (with ventilation as necessary for those time when windows are closed) internal noise levels reduce to 24 and 21 dB respectively irrespective of the location of housing relative to the Industrial Estate.

7.6.12 Sound levels reduce by 6 dB per doubling of distance. Therefore to achieve the BS8233 target internal noise level of 30 dB at night time alternative forms of ventilation (including for instance trickle vents / air bricks) could be provided on property façades facing Synergy Health within 30m. This applies potentially only to habitable rooms. Alternative forms of mitigation will include the layout of the development and of the dwellings.

7.6.13 The mitigation strategy may therefore include the following measures.

- External noise levels are found to be acceptable. Shielding rear gardens behind the proposed housing will only improve the noise environment.
- Consider internal housing layouts to avoid lounges / dining rooms fronting Synergy Health within 30m of the facility.
- Consider passive ventilation systems and double glazing for only those residential fronting Synergy Health within 30 m of the facility.
- Consider the layout of housing to avoid direct sight lines onto Synergy Heath from any habitable room.

7.6.14 Following consideration of the above mitigation strategies, while development can take place adjacent to the Industrial Estate, it is clear that physical mitigation on the properties specific to the environment is within a 30 metre zone of the noise source of Synergy Health. Therefore, on a precautionary basis, it is proposed that a buffer zone of 30 metres is put in place where residential facades would be placed and thereby avoiding the need for further mitigation.

Potential Noise from Keypoint

7.6.15 The assessment set out above takes account of the existing noise sources in the area. Further assessment of the operations now and potentially in the future is set out below. These include the current HGV manoeuvring and the potential for rail freight operations (albeit that no rail freight operations appear to have taken place at Keypoint for several years and none are planned). Moreover the potential for rail freight noise is located at a substantial distance from the nearest point of the application site.

7.6.16 The calculated noise levels due to typical activities during the daytime and night-time are presented in below in Figures 7.5-7.8. The screening described is from the existing 4.5 metre acoustic noise bund and the distance calculated reduction is as calculated as follows: 20log(r2/r1). Distances for HGV activity are taken from adjacent boundaries although in reality HGV movements and activities could not take place immediately along the boundary and would in any event be significantly more distant than the 110 metres employed. The Rail Freight activities are taken from the current location of the tracks which is 270 metres from the nearest point corner of the application site and would therefore be significantly more distant for any significant part of the application site It is presumed that daytime receiver height is ground floor windows, whilst first floor windows would be the night time receiver.

7.6.17 It should be noted that the above figures are direct lines of site measurements and do not take into account the substantial additional noise attenuation associated with buildings associated with these activities within the SDC site and indeed other intervening buildings on the Thornhill Road Industrial Estate. This is particularly

pertinent for the rail freight as this is located on the western edge of the SDC-Keypoint site.

Activity	Noise Level at 10m, $dBL_{Aeq, 1h}$	Distance to Receiver, m	Attenuation, dB		Noise Level $dBL_{Aeq, 1h}$ at Western Boundary
			Distance	Screening	
HGV Manoeuvring	66	110	-21	-15	30
HGV Trailer Hitching	73	110	-21	-15	37
Starting Up	52	110	-21	-15	16
Forklift Truck	65	110	-21	-15	29

Figure 7.5: Calculated noise from HGV activity SDC at western boundary, daytime

Activity	Noise Level at 10m, $dBL_{Aeq, 5min}$	Distance to Receiver, m	Attenuation, dB		Noise Level $dBL_{Aeq, 5min}$ at Western Boundary
			Distance	Screening	
HGV Manoeuvring	63	110	-21	-11	31
HGV Trailer Hitching	70	110	-21	-11	38
Starting Up	49	110	-21	-11	17
Forklift Truck	65	110	-21	-11	33

Figure 7.6: Calculated noise from HGV activity SDC at western boundary, night time

Activity	Noise Level at 10m, $dBL_{Aeq, 1h}$	Distance to Receiver, m	Attenuation, dB		Noise Level $dBL_{Aeq, 1h}$ at Western Boundary
			Distance	Screening	

Shunting Wagon	64	270	-28	-15	21
Idling	40	270	-28	-15	0
Loading/Unloading	62	270	-28	-15	19
Manoeuvring	62	270	-28	-15	19

Figure 7.7: Calculated noise from Rail freight activity SDC at western boundary, daytime

Activity	Noise Level at 10m, $dBL_{Aeq, 5min}$	Distance to Receiver, m	Attenuation, dB		Noise Level $dBL_{Aeq, 5min}$ at Western Boundary
			Distance	Screening	
Shunting Wagon	75	270	-28	-7	40
Idling	51	270	-28	-7	16
Loading/Unloading	62	270	-28	-7	27
Manoeuvring	63	270	-28	-7	28

Figure 7.8: Calculated noise from Rail Freight activity SDC at western boundary, night time

7.6.18 As discussed in previous sections the predominant noise on site as measured and presented in figure 2b is general traffic noise. These figures correspond with the day and night time free field noise measurements of 54dB and 51dB respectively which are higher than any of the noise levels either calculated or predicted from the operations of SDC Keypoint. None of the noise levels potentially associated activity generated by SDC Keypoint result in noise levels at the boundary of the present application site that would result in the need for any specific mitigation measures as part of the application proposals.

Proposed Alternative Location for Primary School at South Marston

7.6.19 The amended application proposals include provision for an additional primary school within the South Marston part of the development.

7.6.20 The location for the alternative primary school is situated where it is least likely to be affected by the primary noise source in the area that of traffic from the existing highways in the area. Consideration was given to noise levels appropriate to education facilities (BB93) in the existing ES and have been reviewed in the light of this further assessment work.

7.6.21 Having regard to the previous noise modelling which indicates that the additional school location at South Marston would be likely to fall within NEC Category A, and the more recent noise observations reported here, the appropriate standards for schools (whether internal or external) can be achieved.

7.6.22 The noise levels across the school site will be dependent upon the detailed layout of the school, the location of playing fields and the use of rooms. Specifically the orientation of the school buildings and the inclusion of appropriate glazing will provide further attenuation benefits if and as appropriate.

7.7 RESIDUAL EFFECTS

7.7.1 The assessments completed above have considered the proposed dwellings in relation to the potential effects from industrial developments in the vicinity including Keypoint and the Thornhill Road Industrial Estate (which includes the premises presently occupied by Synergy Health.

7.7.2 Based on recorded levels, albeit that the noise associated with the industrial uses is intermittent) acceptable internal limits can be achieved for those properties closest to the Thornhill Industrial Estate including if necessary through the through the provision of standard mitigation measures that may include double glazing, together with trickle vents / air bricks within the window frame on habitable rooms or by other means of layout and design. The provision of a precautionary 30metre buffer for facades however avoids the need for further mitigation. No residual effects are anticipated.

7.7.3 Operations either now or in the future at SDC Keypoint are distant enough and shielded enough as to not have a material impact on the development or require any mitigation.

Summary of effects

7.7.4 Having regard to the assessment set out above, without mitigation, the location of the nearest residential properties to the Thornhill Road Industrial Estate (including that

property occupied by Synergy Health) would have no more than a minor adverse impact on residents of those closest properties. With the benefit of the mitigation proposed, such impacts would reduce to negligible.

7.7.5 The location of residential development relative to Keypoint would not result in significant effects on the residents of the properties nearest to that development.

7.7.6 The noise levels likely to be experienced at additional location for the primary school, unmitigated, would be unlikely to result in more than negligible-minor effects. Through design and mitigation such effects would reduce further.